A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 28

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** B - Steam washing (hot water / steam jets kill organisms in the sediment)	1
01.2	**1 mark each** 1 mark for each of the 2 points below. **1** Chemicals cause the oil molecules to join together / solidify **1** Producing more solid material that can be collected / lifted more easily than liquid oil **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
01.3	**1 mark each** 1 mark for each of the 2 points below. **1** Air contains about 210 cm^3 of oxygen per litre whereas water at 20 °C holds a maximum of about 6 cm^3 per litre (less than 3% of the amount in air) **1** So a small amount of oxygen-consuming activity (decomposition, respiration) or warming can remove most of the oxygen – many aquatic organisms are near the lower end of their range of tolerance **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **deoxygenation** → loss of dissolved oxygen from the water • **range of tolerance** → the range of conditions in which a species can survive	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 254-262; AQA spec 3.4.3.2.6-3.4.3.2.7.	
02.1	**1 mark each** 1 mark for each of the 5 points below. **1 Terracing:** sloping land divided into narrow level fields held by retaining walls along contours – water slows as it crosses each terrace and infiltrates instead of running off **1 Tied ridging:** on nearly flat land a criss-cross of intersecting ridges retains rainwater, increasing infiltration and preventing runoff **1 Windbreaks:** hedgerows / tree rows reduce wind velocity (for up to 30 times their height) so the wind has less kinetic energy to carry particles – protect bare fields after harvest **1 Mulching:** organic matter on the surface protects soil from raindrop impact and wind – decomposes to humus which binds particles – retains moisture **1 Strip cropping:** strips of different crops harvested at different times (at 90° to the wind) so there is always a crop protecting part of the field – slows runoff **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Terracing** → cutting the slope into level steps • **infiltration** → water soaking into the soil surface • **humus** → dark, decomposed organic matter in soil **Do not credit** • 'the soil loses its nutrients' as a description of erosion - erosion is the physical loss of the soil itself	5
02.2	**1 mark each** 1 mark for each of the 2 points below. **1** No frequent replanting / soil disturbance (ploughing) that breaks up soil structure and exposes particles **1** Permanent root systems and canopy cover protect the soil all year **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
02.3	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Control stocking density so vegetation regrows faster than it is eaten **1** Restrict access to vulnerable areas such as river banks (fencing) **1** Move / remove livestock at high-risk times, eg after heavy rain when soil is wet **1** Rotational grazing to rest pastures **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'the soil loses its nutrients' as a description of erosion - erosion is the physical loss of the soil itself	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 164-166; AQA spec 3.2.5.3.	
03.1	**1 mark each** 1 mark for each of the 2 points below. **1** Cost increases as the percentage prevented increases **1 The relationship is not linear:** cost rises slowly at first then very steeply / exponentially above about 80-90% **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
03.2	**Calculation** 1 mark for the correct answer; working does not have to be shown. **1** 16.0 - 5.0 = 11.0 (£ million) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (£ million) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	1
03.3	**1 mark each** 1 mark for each of the 4 points below. **1 ALARA:** emissions should be As Low As Reasonably Achievable **1** It is rarely practical / affordable to reduce emissions to zero – low-level emissions may have acceptable impacts so an acceptable emission level is set **1 BATNEEC:** Best Available Technology Not Entailing Excessive Cost - choosing the most effective control equipment whose cost is not excessive **1** The factory would install equipment that gives a high level of control (eg 80-90%) rather than paying £16 million for 99% **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4

Do not write outside the box

Question	Marking guidance	Marks
03.4	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Industry / city may have expanded so there are more pollutant sources and total emissions have risen **1** Better understanding of the health / environmental effects so stricter standards are set **1** Improved technology becomes available (BATNEEC changes) / cost of better control falls **1** Pollutant may have accumulated (persistent) so background levels are higher **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **persistent** → does not break down / stays in the environment for a long time	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 244-245; AQA spec 3.4.3.1.	
04.1	**Any 3 of 5** 1 mark each for any 3 of the 5 points below, in any order, maximum 3. **1** Unburnt fuel in vehicle exhaust gases **1** Evaporation from fuel tanks / spillage at filling stations **1** Solvents (paints, cleaning products) **1** Aerosol propellants **1** Gas leaks / fossil fuel exploitation (methane) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
04.2	**1 mark each** 1 mark for each of the 2 points below. **1** Vapours displaced from the vehicle tank are collected, condensed and returned to the main storage tank **1** Vapours are passed through an activated carbon filter where the hydrocarbons adsorb onto the carbon **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **adsorb** → nutrient ions sticking to the surfaces of clay and humus **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2

Do not write outside the box

Question	Marking guidance	Marks
04.3	**1 mark each** 1 mark for each of the 3 points below. **1** A platinum / palladium catalyst on a high surface-area honeycomb in the exhaust pipe **1** Carbon monoxide and hydrocarbons are oxidised to carbon dioxide (and water) **1** NOx are chemically reduced to nitrogen and oxygen **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **catalytic converter** → a catalyst that converts NOx, CO and unburnt hydrocarbons to nitrogen, carbon dioxide and water	3
04.4	**1 mark each** 1 mark for each of the 2 points below. **1** Binds to haemoglobin (more strongly than oxygen) so haemoglobin cannot carry oxygen from the lungs to tissues **1** Brain damage / death if enough haemoglobin is inactivated – low doses worsen chronic heart disease **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 249, 254; AQA spec 3.4.3.2.4-3.4.3.2.5.	
05.1	**1 mark each** 1 mark for each of the 4 points below. **1 Vehicles:** high energy density (200 Wh kg^{-1}) minimises battery mass so range is greater – many cycles and high efficiency **1 Stationary storage:** mass is unimportant so the low energy density of lead-acid does not matter – lowest cost per kWh – well-established, recyclable **1 Lead-acid drawbacks:** fewer cycles and lower efficiency – lithium fire risk needs management **1** Nickel-cadmium largely replaced because of cadmium toxicity and cost **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	4

Do not write outside the box

Question	Marking guidance	Marks
05.2	**Any 2 of 4** 1 mark each for any 2 of the 4 points below, in any order, maximum 2. **1** Recharging speed **1** Self-discharge during non-use **1** Availability of raw materials (lithium, cobalt) **1** Recyclability / disposal **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
05.3	**1 mark each** 1 mark for each of the 2 points below. **1** Parked electric vehicles plugged into the grid form a very large distributed battery **1** At peaks a small proportion of their stored energy is fed back to the grid – surplus grid energy recharges them - avoiding costly, low-efficiency open-cycle gas turbines **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
05.4	**1 mark each** 1 mark for each of the 2 points below. **1** Surplus electricity electrolyses water to hydrogen, which can be converted to methane **1** Methane is fed into the existing natural gas pipe network and storage, so no new distribution system is needed **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 223-224; AQA spec 3.3.4.1.	

Do not write outside the box

Question	Marking guidance	Marks
06.1	**1 mark each** 1 mark for each of the 5 points below. **1** Trays of the same soil (texture, compaction, organic matter, depth) with different percentage vegetation cover (eg 0, 25, 50, 75, 100% grass), replicated **1** Same slope / gradient – simulated rain from a standard height at a standard rate and drop size (eg watering can / sprinkler) for a standard time **1** Collect the soil splashed / washed off (eg onto a sheet or into a trough at the base), dry and weigh it **1** Repeat each treatment several times and calculate means (and standard deviation) **1** Plot soil loss against cover – statistical test (Spearman's rank / t-test) for significance **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **standard deviation** → a measure of the spread of the data about the mean • **Spearman** → a test for correlation between two sets of ranked data • **t-test** → a test for a difference between the means of two normally distributed sets **Do not credit** • 'the result was significant' without comparing the calculated value with the critical value at p = 0.05	5
06.2	**Any 3 of 5** 1 mark each for any 3 of the 5 points below, in any order, maximum 3. **1** Rainfall intensity / raindrop size / height **1** Soil texture, compaction, organic matter, depth **1** Permeability of the material beneath **1** Gradient **1** Root binding **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Permeability** → how easily water passes through / lets water through **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
06.3	**1 mark each** 1 mark for each of the 2 points below. **1** All variables except the independent variable can be standardised in the laboratory – in the field uncontrollable variables (weather) must be measured instead **1** Laboratory conditions may not represent natural conditions so results may not apply directly to real ecosystems **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **valid** → actually measures what it is meant to measure	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 390-393, 419-420; AQA spec 3.7.1, 3.7.2.2.	

Do not write outside the box

Question	Marking guidance	Marks
07.1	**1 mark each** 1 mark for each of the 2 points below. **1** Friction with the prevailing south-westerly winds drags surface water north-eastwards from the tropical Atlantic **1** Water in the NE Atlantic cools, becomes denser (and more saline as ice forms) and sinks, drawing warm surface water in to replace it – the cold water returns southwards at depth (thermohaline circulation) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **thermohaline** → the ocean conveyor driven by temperature and salinity **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
07.2	**1 mark each** 1 mark for each of the 4 points below. **1** Higher temperatures melt land ice on Greenland which flows into the North Atlantic **1** The fresh water dilutes the seawater, reducing its salinity and density **1** Less dense water sinks less easily so the flow rate of the current is reduced **1** Less heat is transported to NW Europe so the UK could become colder (like Moscow / Churchill at the same latitude) even while global temperatures rise **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4
07.3	**1 mark each** 1 mark for each of the 3 points below. **1** Several interconnected systems act at once over different timescales: jet stream changes may raise temperatures, a slowing Conveyor may lower them, greenhouse gases raise them **1** Natural fluctuations (solar output, orbit) can hide or exaggerate anthropogenic change – individual events do not show trends **1** Time delays between cause and effect (oceans take a long time to warm because of volume and heat capacity) – incomplete understanding of the processes **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **anthropogenic** → human-caused / caused by human activity	3
07.4	1 mark for the point below. **1** Argo floats sink to a set depth for about 10 days, then surface and transmit temperature, salinity and position data – the sequence of positions shows current direction and speed (surface currents also by satellites and buoys) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 96-97; AQA spec 3.2.1.2, 3.2.2.3.	

Do not write outside the box

Question	Marking guidance	Marks
08.1	**2 marks per point** 2 marks for each of the 3 points below; award 1 mark where a point is only partly made. **2** Aerodynamic shapes / fairings reduce friction with the air (air resistance) so less energy is needed to propel the vehicle, especially at speed – hydrodynamics for ships **2** Reduced mass (plastics, carbon-fibre composites, high-strength steel alloys, aluminium engine blocks, lighter batteries and magnets, rounded shapes) means less energy to accelerate and less rolling resistance, so lower fuel consumption **2** Conventional friction brakes convert kinetic energy to waste heat – regenerative braking uses a generator to convert it to electricity stored in a battery (or returned to the rail grid) which then propels the vehicle, so less fuel is needed to accelerate again **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	6
08.2	**1 mark each** 1 mark for each of the 2 points below. **1** Pneumatic tyres deform as they rotate – the continual change in shape and air movement inside generates frictional heat (more if under-inflated) **1** Steel wheels do not deform so rolling resistance is very low **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
08.3	**Any 2 of 4** 1 mark each for any 2 of the 4 points below, in any order, maximum 2. **1** More valves per cylinder to remove exhaust gases fully so they do not mix with fresh fuel **1** Better temperature control (thermostatic fan) **1** Electronic ignition control / regular servicing to keep spark timing correct **1** Turbochargers supplying more oxygen **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 225-227; AQA spec 3.3.4.2.1.	
09.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** B - Tropospheric ozone (formed by reactions of NOx in sunlight)	1

Do not write outside the box

Question	Marking guidance	Marks
09.2	**1 mark each** 1 mark for each of the 4 points below. **1 Primary pollutant:** released directly by human activities, eg sulfur dioxide from burning coal / smelting sulfide ores – NOx from high-temperature combustion **1 Secondary pollutant:** produced by chemical reactions between primary pollutants, often with non-pollutants (water, oxygen, sunlight) **1** Sulfur dioxide is oxidised by ozone to sulfur trioxide which dissolves to form sulfuric acid – NOx dissolve to form nitric acid **1** The acids (sulfurous, sulfuric, nitric) are secondary pollutants – ozone itself is a secondary pollutant of NOx photochemistry **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. **Do not credit** • carbon dioxide as the main cause of acid rain - the gases responsible are sulfur dioxide and oxides of nitrogen	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 237-238; AQA spec 3.4.1.	
10.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** No reservoir range = 3.9 - 0.7 = 3.2 m – with reservoir = 2.7 - 1.7 = 1.0 m **1** Reduction = 2.2 m **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (m) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working. • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	2
10.2	**1 mark each** 1 mark for each of the 2 points below. **1** After heavy rainfall surplus water is stored, reducing peak flows / flooding downstream **1** During dry periods the dam is opened to release water and maintain river levels **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	2

Do not write outside the box

Question	Marking guidance	Marks
10.3	**1 mark each** 1 mark for each of the 2 points below. **1** Deforestation / urbanisation reduce infiltration and interception **1** So runoff reaches the river faster after rain and less water is stored to maintain flow in dry periods **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **infiltration** → water soaking into the soil surface **Do not credit** • 'trees give us the oxygen we breathe' as the main impact - credit carbon storage, soil protection, the water cycle and habitat loss • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	2

Do not write outside the box

Question	Marking guidance	Marks
10.4	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Site factors:** topography - narrow exit from a large deep basin so a small dam holds a large volume – impermeable, strong bedrock without faults or seismic activity – large catchment with regular high rainfall and low evaporation – catchment land use with low pollution risk (pesticides, industry) and low soil erosion to avoid sedimentation – forest / vegetation in the flooded area decaying anaerobically releasing methane – access for workers and materials and distance to demand – value of the land lost (urban, conservation, agricultural) - loss-benefit analysis differs between countries • **Alternatives for supply:** rainwater collection, rivers, estuary barrages, desalination (energy-intensive so only where freshwater is inadequate), unexploited aquifers (N Africa, Kenya), inter-basin transfers (China, Australia, Wales-England), artificial aquifer recharge using surplus wet-season water, river-regulation reservoirs, afforestation to even out flow • **Conservation strategies:** low-water appliances, dual-flush toilets, xeriscaping, drip irrigation, grey-water re-use, leak repair (about 20% lost in the UK), metering • Treatment / pollution control allowing re-use of contaminated sources: sewage treatment, buffer strips, mine drainage control • **Evaluation:** reservoirs give large reliable supplies and flood control but flood land, cause habitat loss and methane – conservation is cheap and quick but limited by behaviour – transfers create conflicts between regions – desalination has high energy / carbon cost – combination needed as demand grows • Justified conclusion	9

Do not write outside the box

Question	Marking guidance	Marks
	Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • **sustainability** → meeting present needs without stopping future generations meeting theirs • **impermeable** → does not let water through • **aquifer** → a permeable rock that stores groundwater / underground water store **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on • 'the soil loses its nutrients' as a description of erosion - erosion is the physical loss of the soil itself	
	Model answer - Level 3 (8 marks)	
	A reservoir site is judged first on topography: a narrow exit from a large, deep basin lets a small, cheap dam hold a large volume. The bedrock must be impermeable so water is not lost and strong enough, without faults or seismic activity, to carry the dam. The catchment should be large with regular, high rainfall and low evaporation, and its land uses should not pose a pollution risk from industry or pesticides or cause the soil erosion that would silt the reservoir. Flooding forest or vegetation causes anaerobic decay and methane. Access for workers and materials, distance to the area of demand, and the value of the land lost - urban areas and nature reserves protected, farmland less so - complete a loss-benefit analysis that different countries weigh differently.	
	Supply can also be increased without new reservoirs: rainwater collection in cities, estuary barrages, desalination where freshwater is genuinely inadequate, newly found aquifers in North Africa and Kenya, inter-basin transfers as in China and from Wales to England, artificial recharge of aquifers with wet-season surplus, river-regulation reservoirs and afforestation to even out flow. Demand can be cut by low-water appliances, xeriscaping, drip irrigation, grey-water re-use, metering and repairing the leaks that lose about a fifth of UK supply, and treatment lets contaminated sources be re-used.	
	Reservoirs give large, reliable supplies and flood control but flood land and habitats and release methane; conservation is cheap and quick but limited by behaviour; transfers create conflicts between regions; desalination carries a high energy and carbon cost. As demand grows, the sustainable answer is conservation first, then a combination of sources matched to each region.	
	Weaker answer - Level 2 (4 marks)	
	A reservoir needs a large catchment area with lots of rain, impermeable rock so the water does not drain away, and a narrow valley so the dam is small. It should be near the people who need the water and not flood important land. Other ways to get water include desalination, which is expensive, and collecting rainwater. Water can be conserved by fixing leaks, using water meters and having low-flush toilets. All of these help make water supplies more sustainable.	
	What makes the difference	

Do not write outside the box

Question	Marking guidance	Marks
	• Level 2 lists correct site factors but without the reasons (why impermeable, why the valley shape matters, why land use in the catchment matters, methane from flooded vegetation).	
	• Level 3 covers far more alternatives (aquifer recharge, river regulation, transfers, afforestation) and evaluates each; Level 2 evaluates only desalination.	
	• Level 3 ends with a prioritised judgement; Level 2 ends with 'all of these help'.	
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 122-127; AQA spec 3.2.2.4-3.2.2.5.	

Do not write outside the box

Question	Marking guidance	Marks
11.1	**Mineral resources are non-renewable. Discuss how an understanding of geological processes and the development of new technologies can extend the period for which society can rely on mineral resources.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Why non-renewable:** formed too slowly – geological concentration processes - hydrothermal, metamorphic, Proterozoic sediments, alluvial, evaporites, secondary enrichment, biological sediments • Understanding processes guides exploration (batholith margins, river gravels, ancient seas) and remote sensing / geophysical techniques • Reserves, resource, stock – Lasky's principle – cut-off ore grade and its drivers • **Extraction technologies:** mechanisation, bioleaching, phytomining, polymer and bacterial adsorption, seabed nodules • **Reducing demand:** recycling and its difficulties, Cradle to Cradle, design for disassembly, linear vs circular material flows, dispersal of metals in use • **Environmental limits:** impacts of mining and their control – energy costs rising with lower grades – conflicts	25
	• **Judgement:** technology can extend supplies substantially but only within energy and environmental constraints – circular economy essential *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 130-147, 373-379; AQA spec 3.2.3, 3.6.3-3.6.4.	

Do not write outside the box

Question	Marking guidance	Marks
11.2	**Discuss how knowledge of the properties of pollutants can be used to reduce the severity of their impact on the environment.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Properties:** state of matter, density, persistence, toxicity, specificity, reactivity, adsorption, solubility, bioaccumulation / biomagnification, synergism, mutagenic / carcinogenic / teratogenic action, mobility • **How properties predict behaviour:** dispersal (gases vs solids, density - Bhopal), how far and how long (persistence, mobility), where they concentrate (adsorption on sediments, food chains) • Environmental features interacting with properties: temperature inversions, currents, pH, oxygen, light, adsorbent materials • **Control based on properties:** persistent / liposoluble pesticides banned (organochlorines) and replaced by non-persistent / specific ones – systemic pesticides and seed coatings • **Heavy metals:** lead uses discontinued, storage at high pH to reduce solubility – mercury chemical form – radioactive waste categorised by half-life / activity and contained (encapsulation, sealed storage) • Reactive primary pollutants controlled to prevent secondary pollutants (catalytic converters, FGD, urea sprays) – oil properties (floats, insoluble) exploited by booms and skimmers	25

Do not write outside the box

Question	Marking guidance	Marks
	– density in cyclone separators • **Principles:** CPA and CGM based on properties – emission location / timing – ALARA and BATNEEC – polluter pays – precautionary principle • **New technologies:** satellite oil-spill monitoring, GPS, polymer adsorption, phytoremediation, bioremediation • **Evaluation:** prevention better than cure – limits of remediation – cost-effectiveness – contrast success of ozone (Montreal) with climate change – judgement *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 236-245, 262-266, 275-285, 295-304; AQA spec 3.4.1-3.4.3.	
	Model answer - Level 5 (22 marks)	
	Pollutants cause damage in proportion to how much reaches sensitive organisms and how harmful it is when it gets there. Both depend on properties, so knowing a pollutant's properties lets us predict where it will go and which control will work.	
	State of matter and density decide dispersal. Solid particulates settle near the source unless they are very small: PM10 and smaller stay suspended, travel far and penetrate lungs, so smoke is controlled at the point of release with electrostatic precipitators, cyclones and bag filters, and smokeless zones prevent its production. Gases disperse widely, and dense gases such as the hydrogen cyanide released at Bhopal hug the ground, so location and emission height matter. Persistence decides whether dilution is acceptable. Sewage and other biodegradable wastes break down, so treatment plus dilution in a river or the sea is sufficient; organochlorine pesticides, CFCs, heavy metals and long-lived radioisotopes are not broken down, so they travel, accumulate and must be contained, banned or replaced - DDT and CFCs were both controlled by substitution once their persistence was understood.	
	Solubility governs the pathway. Water-soluble nitrate leaches into rivers and aquifers, so it is controlled by nitrate vulnerable zones, fertiliser timing and buffer strips; liposoluble pollutants such as DDT and methyl mercury pass into fat and are biomagnified along food chains, as at Minamata, so even low concentrations in water are dangerous and the only effective control is to stop release. Adsorption on clay immobilises caesium and metals, which is why radioactive discharges are monitored in sediments and why activated carbon and polymer adsorbents are used deliberately in effluent treatment.	

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	Toxicity, specificity and mode of action set acceptable doses and identify species at risk. Neonicotinoids are toxic to insects but not vertebrates because they cannot cross the vertebrate blood-brain barrier, which explains both their popularity and their effect on bees; pyrethroids replaced organochlorines because they are less persistent. For mutagens and carcinogens such as ionising radiation there is no safe dose, so exposure is kept As Low As Reasonably Achievable with shielding, distance, time limits and monitoring. Reactivity explains secondary pollutants - sulfur dioxide oxidised to sulfuric acid, nitrogen oxides and hydrocarbons forming ozone and PANs in photochemical smog - so control targets the primary emissions with flue-gas desulfurisation and catalytic converters. Synergism between smoke and sulfur dioxide, or ozone and sulfur dioxide, means limits must consider pollutants together.	
	Properties interact with the environment: temperature inversions trap pollutants, currents disperse them, pH and oxygen alter solubility. Critical Pathway Analysis combines the two to predict where a discharge will concentrate and to focus monitoring, and Critical Group Monitoring checks the most exposed people. These principles have reduced smoke, sulfur dioxide, lead and organochlorines dramatically in developed countries. Their limits are pollutants with diffuse sources or that underpin economies - nitrate, carbon dioxide - where knowing the properties is not the problem; changing behaviour is. Even there, the precautionary principle applies the same logic: predict the properties of a new substance from similar ones and control it before release, because prevention is always cheaper than remediation.	
	What makes the difference	
	• Every paragraph is organised around a **property**, each linked to a prediction and a named control - the essay answers the question throughout rather than describing pollutants.	
	• Examples span the specification (smoke, acid rain, pesticides, heavy metals, radiation, nutrients, oil), with quantitative and named detail (Bhopal, Minamata, PM10, ALARA), which is what Level 5 breadth means.	
	• The final paragraph evaluates: where the approach has worked, where it has not and why, and reaches a justified conclusion.	
	• A Level 3 version of this essay would describe several pollutants and their effects accurately but only occasionally connect a property to a control, and would end without a judgement.	

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